

From a Free-Boundary MOTS Saddle-Node to a Spacelike Marginal Tube in Dynamical Five-Dimensional Gravity

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Abstract

Pair creation is a local statement about marginally outer trapped surfaces (MOTSs); its physical significance depends on what the stable branch becomes. We resolve both the critical creation and the subsequent geometric development of brane-terminating MOTSs in the $SO(3)$ -symmetric sector of a dynamically evolved, Goldberger–Wise-stabilized five-dimensional spacetime. An independently constructed parent sequence contains no accepted capped MOTS at $t/\ell = 0.001$ and exactly two at $t/\ell = 0.0015$ on three successive spacetime grids. Pseudo-arclength continuation locates a common critical time $t_*/\ell \simeq 1.17254 \times 10^{-3}$ in the chosen foliation. The principal stability eigenvalue crosses zero through a real sign-definite mode; discrete adjoint projections give nonzero transversality and quadratic coefficients; and the invariant one-sided proper-area separation follows the predicted square-root law. An independent G10 half-timestep evolution reproduces the critical bracket and gives a free exponent $p \simeq 0.498$. We then follow the stable outer branch over nine evenly spaced matched leaves spanning $0.00121875 \leq t/\ell \leq 0.00146875$, with physical leaf spacing $3.125 \times 10^{-5}\ell$. It forms a resolved spacelike marginal tube with negative inward expansion, strictly increasing area, positive principal stability, and orthogonal brane contact. Its geometry and causal signature transfer across G9/G10/G11 and survive the half-timestep evolution. A brane-inclusive reflected quasi-local charge ledger closes over a finite segment with a worst charge–flux discrepancy of 0.302%, robust under three spatial stencils. The combined result traces a boundary-created stable branch from its saddle-node through a finite-resolution free-boundary dynamical-horizon segment. It remains quasi-local and foliation-dependent and is neither a continuum theorem nor an event-horizon, global-topology, or source-ownership reconstruction.

1 Introduction

Black objects in compact warped geometries probe localization, stabilization, and boundary dynamics simultaneously [1, 2, 3, 4]. Static braneworld calculations have constructed localized solutions from small to large horizon radii [5, 6]; time-symmetric data have produced cigar-shaped apparent horizons, and full Randall–Sundrum II collapse has produced brane-intersecting horizons of finite bulk extent [7, 8]. These results establish that a brane need not confine the relevant horizon geometry to the induced four-dimensional spacetime. They do not, however, determine which free-boundary marginal surfaces are dynamically selected in a stabilized compact two-brane evolution.

Marginally outer trapped surfaces are quasi-local and slicing-dependent, but their defining equation is geometric on each slice and their stability operator organizes local continuation of branches [9, 10, 11, 12]. Multiple MOTSs, branch creation, and alternating stability are established features of four-dimensional black-hole evolutions [13, 14, 15, 16]. Higher-dimensional MOTSs and

free-boundary or capillary MOTSs have independent mathematical treatments, including higher-dimensional stability, curvature, and rigidity results [17, 18, 19, 20, 21, 22]. In that literature, orthogonal contact with the support hypersurface is the free-boundary condition; the brane endpoint condition used below is precisely this orthogonality condition. Related recent work on free-boundary minimal annuli identifies a fold of a static geometric family together with Jacobi–Robin degeneracy [23]. That result is an important neighboring realization, but it concerns minimal surfaces in spherical caps rather than marginal surfaces in an evolved spacetime and does not evaluate the adjoint transversality and quadratic normal-form coefficients tested here. Recent bifurcation analysis identifies generic pair creation as a saddle-node mechanism and distinguishes it from symmetry-supported pitchfork or transcritical cases [24, 25]. The fold mechanism itself is therefore not new. The first question here is whether its hypotheses are realized when the marginal surface has a physical boundary, the contact condition enters the operator domain, and the ambient geometry is supplied by a nonlinear evolution rather than a prescribed metric family. The second is physical rather than classificatory: if a stable outer branch is created, does it persist as a coherent marginal tube with the sign, causal character, area evolution, and balance behavior expected of a dynamical horizon? A spacelike hypersurface foliated by marginal leaves with negative inward expansion is the quasi-local dynamical-horizon structure that captures black-hole growth without reconstructing an event horizon [26, 27].

We answer both questions for brane-terminating free-boundary MOTSs in a compact, explicitly stabilized braneworld. The spacetime is generated by a nonlinear five-dimensional initial-boundary-value evolution; the stability operator is non-self-adjoint; and its domain contains the Robin condition induced by the brane geometry. Existing closed-MOTS bifurcation examples do not contain these features, while existing free-boundary stability and rigidity results do not guarantee dynamical pair creation or the post-fold development of a spacelike tube. The contribution is thus neither a new universal bifurcation class nor pair detection in another metric. It is a controlled numerical connection between a boundary-sensitive fold and the subsequent marginal geometry selected by its stable branch.

Direct three-grid formation, pseudo-arclength continuation, critical-mode and discrete-adjoint tests, invariant square-root scaling, and an independent G10 half-timestep evolution establish the local saddle-node at finite resolution. The stable outer branch is then followed over a fixed post-fold interval. Negative inward expansion, increasing area, positive principal stability, and a uniformly spacelike tube tangent persist under G9/G10/G11 spatial transfer and the half-timestep control. A brane-inclusive local and finite-segment quasi-local balance supplies an independent geometric check. Together these results support a finite-resolution free-boundary dynamical-horizon segment in the imposed $SO(3)$ -symmetric sector and chosen foliation. The analysis does not establish an event horizon, global topology, source ownership, the nonsymmetric nonprincipal spectrum, or a continuum theorem.

Further definitions, prespecified acceptance criteria, extended numerical controls, negative studies, and the reproducibility mapping are given in the Supplemental Material [28].

2 Model and variational convention

2.1 Doubled cover and interval evolution

We use five-dimensional Einstein gravity with $\Lambda_5 = -6/\ell^2$ on the doubled $S^1/\mathbb{Z}_2$ cover. The numerical domain is one fundamental interval. With the Goldberger–Wise field Φ and collapse field

Table 1: Dimensionless model and primary numerical parameters.

quantity	value
ϵ , backreaction, $m_\Phi^2 \ell^2$	0.1, 0.01, 0.41
wall stiffness $\gamma \ell$	20
$m_\chi^2 \ell^2$	0
A , σ_r/ℓ , σ_y , y_c/d	7.90, 1, 0.2, 0.9
repaired critical-sequence domain	$R_{\max}/\ell = 10$
repaired critical grids G9/G10/G11	113×211 , 129×241 , 145×271
repaired evolution step	$\Delta t/\ell = 3.125 \times 10^{-5}$
independent G10 temporal check	$\Delta t/\ell = 1.5625 \times 10^{-5}$

χ , the covering-space action is

$$\begin{aligned}
 S = & \frac{1}{2\kappa_5^2} \int_{\widetilde{\mathcal{M}}} d^5x \sqrt{-g} (R - 2\Lambda_5) \\
 & - \frac{1}{2} \int_{\widetilde{\mathcal{M}}} d^5x \sqrt{-g} [(\nabla\Phi)^2 + m_\Phi^2 \Phi^2] \\
 & - \frac{1}{2} \int_{\widetilde{\mathcal{M}}} d^5x \sqrt{-g} [(\nabla\chi)^2 + m_\chi^2 \chi^2] \\
 & - \sum_i \int_{\Sigma_i} d^4x \sqrt{-h_i} [\lambda_i + U_i(\Phi)], \\
 U_i(\Phi) = & \frac{\gamma_i}{2} (\Phi - v_i)^2.
 \end{aligned} \tag{1}$$

The branes are thin shells on the doubled cover; they are not ordinary outer boundaries of the action in Eq. (1). The evolution is performed on one side of each $\mathbb{Z}_2$ fixed surface. With outward normals from the fundamental interval, the one-sided junction conditions are

$$K_{\mu\nu}^{(B)} = \frac{\kappa_5^2}{2} \left(S_{\mu\nu} - \frac{1}{3} S h_{\mu\nu} \right), \quad n_{\text{out}}^a \nabla_a \Phi = -\frac{1}{2} U'_i(\Phi), \tag{2}$$

where $K_{\mu\nu}^{(B)}$ is the brane extrinsic curvature. An equivalent one-sided interval variational principle contains the appropriate Gibbons–Hawking–York terms and doubles the one-sided bulk contribution relative to each brane action [29, 30]. The thin-shell normalization follows the Israel junction formalism [31]. The junction conditions in Eq. (2) reproduce the tuned Randall–Sundrum signs at both walls. This is the convention implemented in the initial-data and evolution boundary rows.

2.2 Normalization and initial family

All reported quantities are nondimensionalized by the AdS radius. We set $\ell = \kappa_5 = 1$, take $d/\ell = 1$, and use

$$z \in [1, e], \quad r/\ell \in [0, R_{\max}/\ell]. \tag{3}$$

For the load-bearing repaired G9/G10/G11 line, $R_{\max}/\ell = 10$. The superseded pre-repair G7/G8 line retained only as a historical record in the supplement used $R_{\max}/\ell = 8$. Thus the quoted time is t/ℓ . The background parameters and primary discretizations are listed in Table 1.

Writing $y = \log z$ and $d = 1$, the collapse field is initialized as

$$\chi(z, r) = A P_N(y; y_c, \sigma_y, d) \exp\left(-\frac{r^2}{2\sigma_r^2}\right), \quad (4)$$

where P_N is the normalized even periodic-image sum that enforces Neumann data at both compact walls. Five image pairs are retained. Therefore $A = 7.90$ in Eq. (4) is the peak coefficient of a completely specified pulse family, not an invariant critical parameter. Initial data are momentarily time symmetric,

$$K_{ij} = 0, \quad \Pi_\Phi = 0, \quad \Pi_\chi = 0, \quad (5)$$

and the coupled Hamiltonian/stabilizer system is solved before evolution.

3 Capped MOTSs and the free-boundary stability problem

Let u^a be the future unit normal to a spatial slice M , and let N^a be the outward unit normal to a candidate three-surface $\mathcal{S} \subset M$. For $\ell_+^a = u^a + N^a$, our outgoing expansion is

$$\theta_+ = D_i N^i + K_{ij} N^i N^j - K. \quad (6)$$

A MOTS satisfies $\theta_+ = 0$ in Eq. (6). The upper-brane-capped class consists of $SO(3)$ -symmetric polar graphs

$$z = z_B - \rho(\vartheta) \cos \vartheta, \quad r = \rho(\vartheta) \sin \vartheta, \quad 0 \leq \vartheta \leq \frac{\pi}{2}, \quad (7)$$

with regular closure at the axis and orthogonal contact with the upper brane.

3.1 Physical Robin condition and its coordinate reduction

The free-boundary perturbation condition is derived rather than imposed as a coordinate endpoint rule. Let ϱ be the outward unit normal to the brane within M , let ν be the outward conormal to $\partial\mathcal{S} \subset \mathcal{S}$, and let N be the unit surface normal. At orthogonal contact,

$$\nu = \varrho, \quad g(N, \varrho) = 0. \quad (8)$$

For a variation with physical normal component fN , linearizing the contact condition gives the standard free-boundary operator

$$Bf := \nabla_\nu f - \Pi^{\partial M}(N, N)f = 0 \quad \text{on } \partial\mathcal{S}, \quad (9)$$

where $\Pi^{\partial M}(X, Y) = g(\nabla_X \varrho, Y)$ [18, 20]. The brane curvature term is therefore not optional.

It is nevertheless already contained in the graph variable used by the calculation. The upper-brane cap terminates at the upper wall $z = e$. We orient the wall normal ϱ out of the retained interval, the surface conormal ν outward from $\mathcal{S}$, and the MOTS normal N toward increasing areal radius. Wall parity gives $h_{zr} = 0$ there, and hence $N = h_{rr}^{-1/2} \partial_r$ and $\nu = \varrho = h_{zz}^{-1/2} \partial_z$ at the cap endpoint. A graph displacement $\delta\rho$ has physical normal amplitude

$$f = w \delta\rho, \quad w = g(N, \partial_r) = \sqrt{h_{rr}}. \quad (10)$$

The same wall geometry gives

$$\Pi^{\partial M}(N, N) = \frac{1}{2} \nu(\log h_{rr}) = \nu(\log w). \quad (11)$$

Substituting Eq. (11) into Eq. (9) yields

$$Bf = w \nu(\delta\rho). \quad (12)$$

Consequently, the implemented endpoint condition $\partial_\vartheta \delta\rho = 0$, equivalently $\partial_\vartheta(f/w) = 0$, is the coordinate form of the physical Robin condition for the wall-orthogonal graphs in Eq. (7); it does not discard the brane-extrinsic-curvature term. The same zero derivative at $\vartheta = 0$ is the smooth-axis regularity condition.

The stability operator is

$$Lf = \delta_{fN}\theta_+, \quad Bf = 0. \quad (13)$$

The problem in Eq. (13) is evaluated on the complete evolved slice and is generally non-self-adjoint [11]. With our sign convention, a nonnegative principal eigenvalue is outward-stable. For the connected cap, the scalar stability operator is strongly elliptic, and its Robin domain commutes with the $SO(3)$ action. Free-boundary principal-eigenvalue theory supplies a positive eigenfunction that is unique up to scale [18, 22]; applying any group element therefore reproduces the same normalized eigenfunction. It follows that the principal eigenfunction is $SO(3)$ -invariant and the symmetric calculation captures the full principal eigenvalue and its simple kernel. This argument does not determine the nonprincipal nonsymmetric spectrum. The matrix is a centered physical-normal Fréchet linearization after seven-point local finite-difference elimination of the endpoint Robin conditions. The boundary treatment is checked by the manufactured Robin spectrum, endpoint defects below 10^{-10} , and 49/65/81 angular-node sequences; no alternative stencil-width study is claimed. The boundary derivation, endpoint conventions, and manufactured Robin control are documented in the Supplemental Material [28].

3.2 Boundary-value map and local normal form

The nonlinear problem is not merely the scalar equation $\theta_+ = 0$ on an unrestricted profile space. Before endpoint elimination it is the boundary-value map

$$\begin{aligned} \widehat{\mathcal{F}}(\rho, t) &= (\theta_+, \mathcal{R}_0, \mathcal{C}_B) [\rho; t] = 0, \\ \mathcal{R}_0 &= \rho'(0), \quad \mathcal{C}_B = g(N, \varrho)|_{\vartheta=\pi/2}, \end{aligned} \quad (14)$$

where $\mathcal{R}_0 = 0$ enforces smooth-axis regularity and $\mathcal{C}_B = 0$ enforces orthogonal brane contact. Linearization at a solution gives the interior stability operator L , the axis regularity row, and the physical Robin row B in Eq. (9). Thus the brane geometry enters the continuum Robin domain and, after endpoint elimination, the computed discrete kernel and discrete adjoint projection; it is not an external endpoint check applied after the stability calculation.

Near a critical solution (ρ_*, t_*) , the coordinate reduction above supplies a local chart of admissible surfaces, and the discretization eliminates the endpoint rows. We use the physical normal amplitude f in that chart and write the reduced interior residual as $\mathcal{F}(f, t)$, with $\mathcal{F}(0, t_*) = 0$. Its derivative with respect to f is the stability operator L on the Robin domain $Bf = 0$. This is also what the numerical endpoint elimination produces. For this finite-dimensional reduced map, let f_R span the simple right nullspace of $D_f \mathcal{F}$ and let f_L span its one-dimensional Euclidean left nullspace. With $\langle f_L, f_R \rangle = 1$, write

$$f = s f_R + \eta(s, \tau), \quad \tau = t - t_*, \quad \langle f_L, \eta \rangle = 0. \quad (15)$$

Solving the range equation for η and projecting the remaining equation onto f_L gives the local

scalar reduction

$$\begin{aligned}
0 &= a\tau + bs^2 + O(\tau^2 + |\tau s| + |s|^3), \\
a &= \langle f_L, \partial_t \mathcal{F} \rangle, \\
\mathcal{H} &= D_f^2 \mathcal{F}(0, t_*), \\
b &= \frac{1}{2} \langle f_L, \mathcal{H}[f_R, f_R] \rangle.
\end{aligned} \tag{16}$$

For $a \neq 0$ and $b \neq 0$, on the side where $-a\tau/b > 0$, the two branches satisfy

$$s_{\pm} = \pm \sqrt{-\frac{a}{b}\tau} + O(\tau). \tag{17}$$

Consequently, any smooth, reparameterization-invariant geometric surface functional Q with $DQ[f_R] \neq 0$ separates as

$$Q_+ - Q_- = 2DQ[f_R] \sqrt{-\frac{a}{b}\tau} + O(\tau). \tag{18}$$

The calculation below tests these logically distinct ingredients with the principal zero mode, the discrete adjoint coefficients a and b , and the one-sided proper area $Q = \mathcal{A}$. Equation (16) and its branch and invariant consequences in Eqs. (17) and (18) state the local numerical hypothesis being tested; they are not asserted here as a new continuum theorem [32, 24].

3.3 Post-fold marginal-tube observables

Let u^a be the future unit normal to a spatial slice, s^a the outward unit normal to a marginal leaf, and $\ell^a = (u^a + s^a)/\sqrt{2}$, $n^a = (u^a - s^a)/\sqrt{2}$. The normal projection of the positive-time tube generator is written

$$X^a = A\ell^a - Bn^a, \quad X^2 = 2AB. \tag{19}$$

A future-outward spacelike segment requires $A > 0$ and $B > 0$, not merely $X^2 > 0$. On a marginal leaf $\theta_{(\ell)} = 0$, while $\theta_{(n)} < 0$ supplies the future trapped sign. If the full deformation is $\mathcal{X}^a = X^a + V^a$, with V^a tangent to the leaf, orthogonal brane contact and endpoint motion within the brane give $V \cdot \nu = 0$. The one-sided area law is therefore

$$\frac{d\mathcal{A}}{d\lambda} = - \int_S B\theta_{(n)} dA > 0. \tag{20}$$

The physical cap is open, so its intrinsic-curvature balance is evaluated on the reflected three-leaf. The resulting AdS-subtracted quasi-local charge candidate Q_{AdS} contains a coupled cap-brane seam contribution fixed by the Israel geometry. The finite-segment test has the form

$$Q_{\text{AdS}}(\lambda_2) - Q_{\text{AdS}}(\lambda_1) = \int_{\lambda_1}^{\lambda_2} F_{\text{bulk+brane}} d\lambda. \tag{21}$$

This is a quasi-local balance on the tested tube segment, not a global mass or source-ownership ledger. The boundary variation, charge convention, and separate spatial and temporal tests are given in the Supplemental Material [28].

4 Numerical strategy

The coupled Einstein–scalar equations are evolved in a regular $SO(3)$ -reduced generalized-harmonic formulation [33]. Compact-wall rows own the Israel and scalar conditions, interior-axis rows own regularity, and the wall–axis and wall–outer corners use a fixed staged ownership policy. A second wall-owner policy is retained as a control. The surface analysis is performed only after finite-value, Lorentzian-signature, constraint, wall/axis, and endpoint checks pass.

The load-bearing repaired-line detector solves the local second-order cap equation as an adaptive two-point boundary-value problem with $\rho'(0) = \rho'(\pi/2) = 0$ from the fixed seed bank

$$1.15, 1.20, \dots, 1.70. \quad (22)$$

Admission uses the local residual, endpoint slope, physical domain, and an independent expansion cross-evaluator. The 10^5 condition-number threshold reported in the supplement belongs only to a spectral detector used on the superseded pre-repair line. It is not applied to the repaired direct search, the local BVP continuation, or the augmented pseudo-arclength corrector. The principal stability operator whose eigenvalue crosses zero is likewise not that spectral detector matrix. In particular, singularity of the stability operator at the fold does not imply that the unrelated historical detector gate is triggered. No continuation condition-number trace was archived, so no conditioning trend is asserted or used as evidence.

The load-bearing critical calculation uses an independently constructed, parity-repaired parent sequence on G9, G10, and G11. A direct search is empty at $t/\ell = 0.001$ and admits exactly two branches at $t/\ell = 0.0015$ on all three grids. From the outer branch at the latter time, the local cap equation is continued backward with time promoted to an unknown and an integral pseudo-arclength condition appended [34]. The spacetime between stored RK2 endpoints is supplied by the fixed second-order dense extension of the evolution method; every endpoint replays bitwise. Corrector failures trigger only a prespecified dyadic arclength backoff.

After the principal eigenvalue changes sign, the arclength interval is halved until the time bracket is below $\Delta t/64$. At the smaller-magnitude endpoint of the final bracket, the physical-normal stability matrix is evaluated at 49, 65, and 81 angular nodes and repeated at a doubled Fréchet step. Because the operator is non-self-adjoint, its discrete left and right modes require an explicit convention. On the interior uniform angular nodes, the right eigenvector is phased so that its largest component is positive real and scaled to unit ℓ_∞ norm; the Euclidean conjugate-transpose left eigenvector is then scaled so that $f_L^* f_R = 1$. No mass matrix or continuum quadrature enters this normalization. The reported coefficient magnitudes are therefore reproducible near-critical discrete convention values; their resolved nonvanishing, rather than invariance under rescaling, is the normal-form test. Any admissible change of eigenvector normalization rescales a and b by finite nonzero factors and therefore cannot change whether either coefficient vanishes. After endpoint elimination, the two coefficients in Eq. (16) are evaluated as

$$a = \langle f_L, \partial_t \theta_+ \rangle, \quad b = \frac{1}{2} \langle f_L, \partial_{f_R}^2 \theta_+ \rangle, \quad (23)$$

with centered step-halving controls; these are the standard fold nondegeneracy quantities [32]. Finally, both branches are independently resolved at five prespecified time offsets and $(\mathcal{A}_{\text{out}} - \mathcal{A}_{\text{in}})^2$ is fitted linearly in time.

The load-bearing initial statement is explicitly search-bounded. On the G9 and G10 constraint-solved initial slices, regular-axis shooting uses 2,049 uniform launch radii covering

$$0.10 \leq \rho_0/\ell \leq 1.67, \quad (24)$$

Table 2: Direct three-grid pair geometry and stability at $t/\ell = 0.0015$. $\mathcal{A}$ is the one-sided proper cap area.

grid	$\mathcal{A}_{\text{in}}$	λ_{in}	$\mathcal{A}_{\text{out}}$	λ_{out}
G9	40.67520320	-0.25894888	40.72845794	+0.29347891
G10	40.67520099	-0.25895908	40.72845823	+0.29348687
G11	40.67519977	-0.25896247	40.72845818	+0.29348984

together with twelve cutoff, tolerance, and step variants. Opposite-wall caps, closed star-shaped surfaces, and interval-spanning surfaces are searched separately in their frozen bounded classes. The separate $A = 7.94$ positive control belongs to the superseded pre-repair study and is not used to qualify this initial null result. The complete launch, seed, surface-class, and ordered-gate inventories are given in the Supplemental Material [28].

AI use in the numerical work. OpenAI ChatGPT 5.3 and OpenAI Codex (GPT-5 series, accessed August 2026) assisted with code drafting and debugging, prospective protocol development, and the organization of numerical audits. The author specified the equations, acceptance criteria, and scientific interpretations; reviewed the generated code; and verified the reported outputs through preserved replay tests, independent implementations, spatial refinement, and the half-timestep calculation. The tools did not autonomously alter archived data or select post hoc acceptance thresholds.

5 Results

5.1 Three-grid zero-to-two formation

The independently constructed repaired critical sequence contains no accepted upper-brane-capped branch at $t/\ell = 0.001$ and exactly two at $t/\ell = 0.0015$ on G9, G10, and G11:

$$0.001 < t_{\text{form}}/\ell \leq 0.0015. \quad (25)$$

Equation (25) is a sampled formation bracket, not the later continuation estimate. Each branch is recovered from six seeds. Table 2 shows the direct geometric and stability observables. The adjacent-grid relative differences are at most 1.02×10^{-6} for either endpoint, 1.76×10^{-7} for the proper geometry, and 3.94×10^{-5} for the principal eigenvalue. The negative inner and positive outer eigenvalues therefore close directly across three spacetime grids, rather than being inferred from convergence of the underlying fields.

5.2 Continuation to the critical surface

Backward pseudo-arclength continuation follows the stable outer branch around the turn and onto the unstable inner branch. It encloses the principal zero in time intervals of width 1.84×10^{-7} (G9), 1.72×10^{-7} (G10), and 1.87×10^{-7} (G11). To the precision supported by the temporal check below, each gives

$$\frac{t_*}{\ell} \simeq 1.17254 \times 10^{-3}. \quad (26)$$

The gridwise values summarized in Eq. (26) have full spread 9.23×10^{-9} in time and 8.55×10^{-5} in the selected near-critical one-sided area. Both sequences are nonmonotone, as are the coefficient sequences

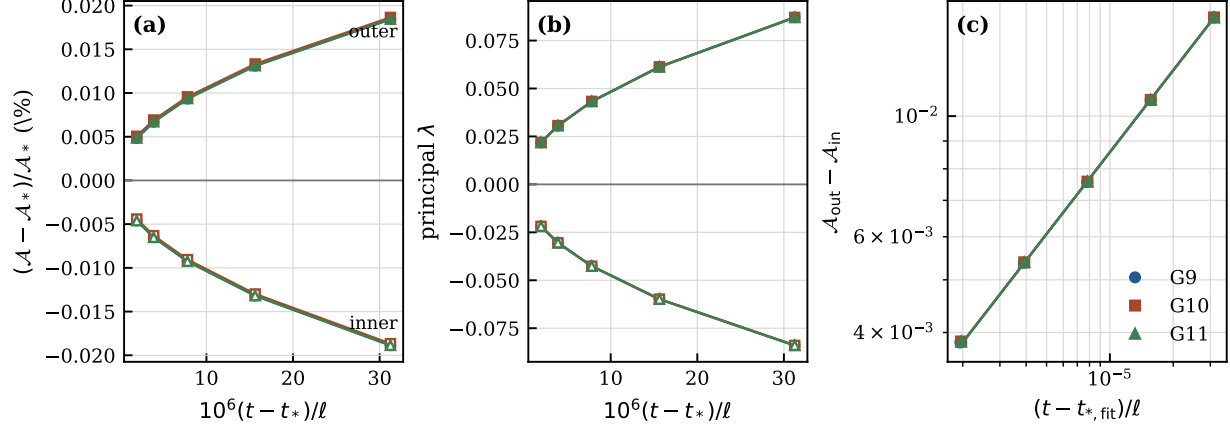


Figure 1: Three-grid resolution of the free-boundary MOTS saddle-node. **(a)** One-sided proper cap area relative to the gridwise near-critical reference area $\mathcal{A}_*$ at the selected bracket endpoint, not an exact critical value; filled symbols denote the outer branch and open symbols the inner branch. **(b)** The corresponding principal stability eigenvalues, which approach zero with opposite signs. **(c)** Log-log area separation and the gridwise fits $\Delta\mathcal{A} = [s(t - t_{*,\text{fit}})]^{1/2}$.

Table 3: G10 half-timestep comparison. The t_* entries are bracket midpoints; their difference is not an uncertainty estimate. Differences for $\mathcal{A}_{\text{nc}}$, a , and b are relative; $\mathcal{A}_{\text{nc}}$ is the selected near-critical reference area.

quantity	$\Delta t/\ell$	$\Delta t/(2\ell)$	difference
t_*/ℓ	0.0011725452	0.0011725443	9.17×10^{-10}
$\mathcal{A}_{\text{nc}}$	40.703366	40.703400	8.45×10^{-7}
a	-20.76565	-20.76535	1.42×10^{-5}
b	2.80410	2.80435	9.23×10^{-5}
free exponent p	0.49597	0.49806	—
R^2 for squared-area fit	0.9999804	0.9999951	—
zero-bracket width	1.72×10^{-7}	1.79×10^{-7}	—

below, so they are reported as finite-resolution scatter rather than assigned a Richardson order or a continuum extrapolation. The representative near-critical one-sided cap area is $\mathcal{A} \simeq 40.7034$.

Figure 1 displays the resolved branch structure. Proper area and principal eigenvalue separate with opposite signs on the two sides, while the area separation follows the square-root law on every grid.

5.3 Independent half-timestep comparison

The spatial sequence shares a second-order RK2 step and dense extension, so its grid agreement alone cannot bound a common temporal bias. We therefore reconstructed the G10 parent from $t/\ell = 0.001$ to 0.0015 with $\Delta t/(2\ell) = 1.5625 \times 10^{-5}$, independently re-solved both endpoint anchors, replayed the full half-step trajectory, and repeated the continuation, stability, coefficient, and area-scaling calculations. Table 3 shows the direct comparison. All criteria were fixed before the half-step result was examined.

The full- and half-step critical-time brackets overlap. Their midpoint shift is 9.17×10^{-10} , while

Table 4: Near-critical discrete normal-form observables. Coefficients are evaluated at the smaller-magnitude endpoint of the final zero bracket; the final entries of the two-step evaluations are shown.

quantity	G9	G10	G11
next reduced-sector eigenvalue	5.56598	5.56531	5.56595
near-critical transversality a	−20.76513	−20.76565	−20.76493
near-critical quadratic coefficient b	2.80717	2.80410	2.81060
square-root exponent p	0.4959	0.4960	0.4960
R^2 for $(\Delta\mathcal{A})^2$ fit	0.9999801	0.9999804	0.9999803

the brackets themselves remain approximately 1.8×10^{-7} wide; the midpoint shift is therefore not interpreted as the uncertainty in t_* . The half-step run retains the isolated near-critical principal mode, coefficient signs, and square-root scaling. The measured changes satisfy the prospective ceilings: $\Delta t/64$ for the bracket-midpoint shift, 1% for the near-critical area, and 20% for each coefficient. The prescribed decision rule therefore strongly constrains the shared temporal-discretization concern without triggering a quarter-step calculation. Although this is not a separate convergence study of constraint violation, the complete half-step evolution independently passes the same native constraint and boundary gates while reproducing the critical observables. It therefore supplies an indirect temporal check against systematic evolution error rather than only a continuation-level comparison. The prospective temporal criteria and full comparison record are reported in the Supplemental Material [28].

5.4 Near-critical mode, nondegeneracy, and invariant scaling

Table 4 gives the near-critical diagnostics. The representative profile is the smaller-magnitude endpoint of the final opposite-sign bracket, so its listed eigenvalue need not vanish identically. At those representatives, $|\lambda|$ lies between 7.4×10^{-4} and 1.7×10^{-3} . The next eigenvalue in the $SO(3)$ -reduced discretization remains near 5.566, leaving a large reduced-sector spectral gap. The principal right mode is real and sign-definite. The unit left–right overlap is imposed by the normalization above and is not counted as independent evidence.

The centered step-halving evaluations of Eq. (23) retain $a < 0$ and $b > 0$ on every grid. Adjacent relative differences are at most 3.46×10^{-5} for a and 2.32×10^{-3} for b , compared with the prospective 0.20 ceilings. Together, the isolated near-critical principal mode and its bracketed zero crossing are consistent with a simple critical kernel; the temporal crossing is transverse and the quadratic term is resolved and nonzero under the stated discrete convention. Independently, five paired fixed-time solves per grid give

$$\mathcal{A}_{\text{out}} - \mathcal{A}_{\text{in}} \propto (t - t_*)^p, \quad p \simeq 0.496 \quad (27)$$

Equation (27) holds on the three-grid sequence, and $p \simeq 0.498$ in the half-step G10 run. The quoted $R^2 = 0.9999801\text{--}0.9999804$ belongs to the linear fit of $(\Delta\mathcal{A})^2$ against time, not to the separate free-exponent fit. Fixing t_* to the continuation midpoint gives $p = 0.4951\text{--}0.4964$; omitting one of the five offsets gives $p = 0.4919\text{--}0.4996$. A fixed $p = 1/2$ fit has maximum relative residual 0.9–1.2%. These sensitivity checks support a near-one-half law but do not justify treating the last exponent digits as an uncertainty estimate. The shift from $p \simeq 0.496$ at the full step to $p \simeq 0.498$ at the half step is toward one-half, consistent with a residual temporal-discretization bias rather than a fit-selection artifact; two time steps are not used to infer an error order. The continuation, operator, discrete-adjoint, and invariant-geometry tests therefore identify the local saddle-node structure through mutually distinct observables.

Table 5: Post-fold marginal-tube controls. Relative differences are maxima over the stated comparison unless otherwise noted.

risk	criterion	outcome
grid-dependent tube	G9/G10/G11 geometry < 1%	maximum geometry difference 1.24×10^{-6} ; stability signs agree
unresolved causal character	every surface node space-like	all 501 nodes on every tested interior leaf resolve spacelike
temporal bias	full/half G10 difference < 10^{-2}	maximum causal-signature difference 1.733×10^{-8} ; minimum resolution margin 54.01
incorrect future branch	$\theta_{(n)} < 0$, $\dot{\mathcal{A}} > 0$, $\lambda_0 > 0$	all three signs persist in both timestep lanes
incomplete flux ledger	charge-flux discrepancy < 1%	maximum 0.302%; brane-endpoint subtotal 59.44% of integrated flux

5.5 Post-fold development of the stable branch

The fold calculation alone does not show that neighboring outer MOTSs belong to a physically coherent tube. We therefore follow the accepted stable branch over the fixed interval $0.00121875 \leq t/\ell \leq 0.00146875$. At each time the surface profile, principal stability eigenvalue, both null expansions, and the normal projection of the tube generator are reconstructed independently. Backward, centered, and forward velocity paths are retained so that a positive centered norm is accepted as spacelike only when it is separated from the one-sided spread. The resulting tests are summarized in Table 5. They directly evaluate the oriented generator and area law in Eqs. (19)–(20) and the finite-segment balance in Eq. (21).

Figure 2 makes the nine-leaf sampling and its temporal and spatial comparisons explicit. The largest stored inward expansion is approximately -2.8587×10^{-2} , the marginality defect is at most 1.85×10^{-6} in the interior, and the orthogonal-contact residual is at most 1.10×10^{-18} . The integrated balance is tested on the fixed subsegment $0.001375 \leq t/\ell \leq 0.00146875$. Its worst adjacent-grid transfer metric is 7.58×10^{-5} and its worst within-grid spatial-stencil metric is 3.54×10^{-5} . The brane terms are nonzero and are retained in the ledger rather than absorbed into the residual. As shown in Fig. 3, their subtotal supplies approximately 59.44% of the integrated flux on every grid. The seam coefficient $c = W_s/W$ is fixed from the Israel boundary data and the coupled seam law is derived before the charge-flux residual is examined; neither is fitted to obtain closure.

These outcomes address distinct alternatives: surface persistence alone could hide a timelike or unresolved tube; area increase alone could be a finite-difference artifact; and a small bulk-only residual could omit the moving cap-brane joint. Their joint closure supports the finite-resolution interpretation that the fold-selected stable branch develops into a spacelike free-boundary marginal, or dynamical-horizon, segment. It does not turn the segment into an event horizon or a global conserved-charge construction. Extended values and the boundary-coupled balance derivation are reported in the Supplemental Material [28].

6 Interpretation

The first result is a boundary-sensitive test of the local MOTS bifurcation framework, not simply another observation of pair formation. The physical contact condition is part of the nonlinear map

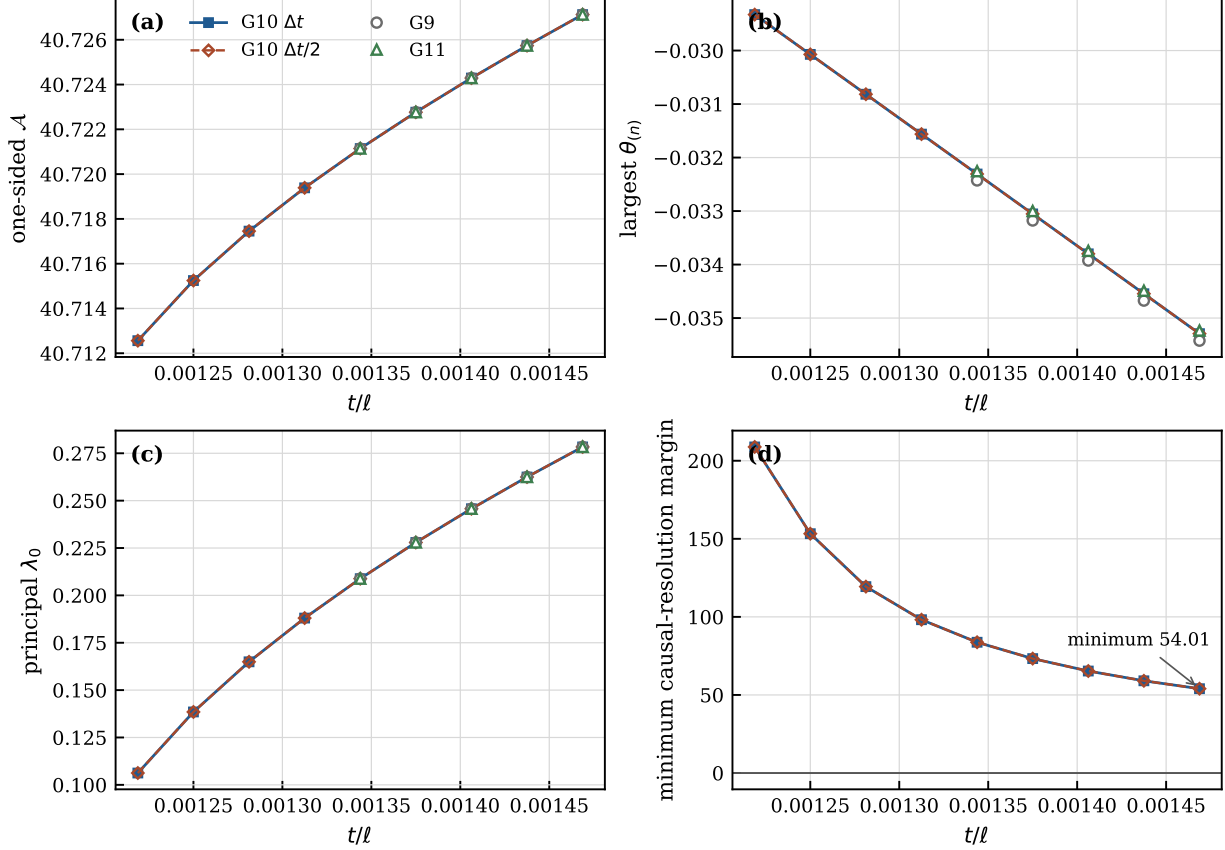


Figure 2: Post-fold evolution of the stable outer branch. The independently evolved G10 full- and half-timestep lanes coincide over the nine matched leaves; open G9 and G11 markers show the common later spatial-transfer samples. The one-sided area increases, the least-negative inward expansion remains below zero, and the principal eigenvalue remains positive. The causal-resolution margin is $\min_{\theta}(|\Xi_c| - \delta\Xi)$, where $\delta\Xi$ is the maximum one-sided stencil spread; it remains positive at every one of 501 nodes and has minimum 54.01 in the fixed generator normalization.

in Eq. (14); its linearization fixes the Robin domain, the endpoint-eliminated discrete kernel, and the corresponding discrete adjoint projection. The branches are connected by pseudo-arclength continuation; an isolated near-critical principal mode accompanies the bracketed zero crossing; both discrete-adjoint nondegeneracy projections are nonzero; and an invariant geometric separation has the expected one-half exponent. None of these ingredients is inferred from the branch count alone. Their agreement across three spacetime grids and the independent half-timestep G10 comparison support a numerical free-boundary MOTS saddle-node at finite resolution in the imposed $SO(3)$ -symmetric problem.

This chain also supplies a portable diagnostic for other physical-boundary MOTS problems: formulate the contact condition as part of the boundary-value map, continue through the singular point, resolve the left and right critical modes, test the two nondegeneracy projections, and confirm the predicted law with an invariant geometric observable. The present calculation establishes that chain only for the stated discretized evolution; it does not prove that every free-boundary MOTS problem has the same unfolding. This single realization establishes non-obstruction in the tested surface class, not genericity across free-boundary MOTS problems.

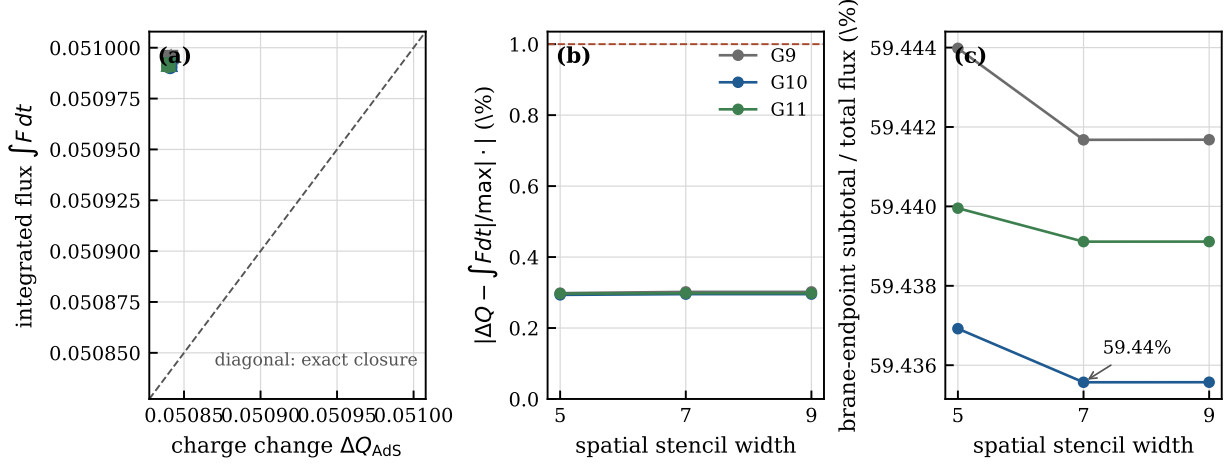


Figure 3: Brane-inclusive balance on the fixed segment $0.001375 \leq t/\ell \leq 0.00146875$. (a) The charge change and integrated flux lie close to the exact-closure diagonal for G9/G10/G11 and stencil widths 5, 7, and 9. (b) Every relative charge–flux discrepancy remains below the prospective 1% ceiling. (c) The coupled brane-endpoint subtotal supplies approximately 59.44% of the integrated flux. This is a local ledger decomposition, not a global source attribution.

The second result supplies the physical continuation. The stable branch does not merely recur on adjacent slices: the resolved normal generator is future-outward and spacelike, the inward expansion is negative, the area increases, and the principal eigenvalue remains positive. The brane-inclusive finite-segment balance then connects the changing quasi-local charge to an explicit bulk-plus-boundary flux ledger. Spatial transfer, stencil variation, and an independently evolved half-timestep parent constrain different numerical explanations for that combined behavior. In this sense the fold selects a stable branch that becomes a finite dynamical-horizon segment rather than an isolated collection of roots.

The concrete significance lies in both the surface class and the dynamical sequence. The MOTSs terminate orthogonally on a brane inside an explicitly stabilized compact interval. They are neither imposed as static horizons nor continued from a pre-existing connected black string. The established saddle-node mechanism is realized with a physical Robin boundary and is then followed into a spatially and temporally controlled free-boundary marginal tube. That connected sequence, rather than the abstract fold mechanism, is the principal result.

7 Limitations and boundary of inference

The initial null result is bounded by the stated launch interval, seed family, and searched surface classes. MOTSs depend on slicing; no repaired-line gauge family is surveyed here. The critical time is a finite-resolution continuation estimate in the chosen foliation. The half-timestep comparison constrains the common temporal bias, but the nonmonotone three-grid scatter is not promoted to a continuum theorem, Richardson order, or formal uncertainty interval. Equivariance and uniqueness place the principal mode in the $SO(3)$ -invariant sector, but the nonprincipal spectral-gap statement and the discrete normal-form evaluation remain tied to the symmetric branch and stated discretization. They are numerical results, not analytic Fredholm bounds on the uncomputed nonsymmetric nonprincipal spectrum. Separately solved larger-domain parents do not define the

same universal formation datum. The marginal-tube claim is restricted to nine matched leaves over the tested finite interval, the same symmetry sector, and the chosen foliation. Its reflected quasi-local charge is a boundary-aware finite-segment diagnostic, not a global mass, source-ownership, or inter-sector transfer ledger. Finally, no event horizon, future-complete causal boundary, common horizon, global topology change, or nonlinear stability theorem follows from the evidence. The archived negative, failed, and incomplete studies underlying these bounds are summarized in the Supplemental Material [28].

These are not withheld conclusions from the same calculation. Each requires a different observable or a different global construction.

8 Conclusion

We have followed a dynamically selected free-boundary marginal geometry from its local creation to a finite spacelike tube segment. The physical contact condition enters the nonlinear MOTS map, its Robin domain, and the endpoint-eliminated discrete adjoint projection. Across G9, G10, and G11, the evolved parent changes from no accepted capped surface to two opposite-stability branches. Pseudo-arclength continuation joins those branches at a grid-consistent critical surface; the principal mode crosses zero; both discrete-adjoint nondegeneracy coefficients are nonzero; and the invariant area separation follows the expected near-one-half law. The independent half-timestep parent reproduces this critical structure.

The stable outer branch then develops into a future-outward spacelike marginal tube with negative inward expansion, increasing area, positive principal stability, orthogonal contact, and a brane-inclusive finite-segment balance. Those properties survive the same three-grid spatial transfer and the independent G10 half-timestep evolution. The combined result therefore moves beyond pair anatomy: it numerically resolves the boundary-sensitive fold and shows what its stable branch becomes over a controlled post-fold interval. This supplies a reusable route from free-boundary MOTS bifurcation to local dynamical-horizon geometry in an evolved higher-dimensional spacetime. The result remains finite resolution and quasi-local; no event horizon, global topology, source ownership, or continuum theorem is inferred.

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Data Availability

Machine-readable result records and frozen protocol manifests are archived in the Zenodo version identified by [doi:10.5281/zenodo.22148554](https://doi.org/10.5281/zenodo.22148554). The complete version series is identified by the concept DOI [doi:10.5281/zenodo.22085054](https://doi.org/10.5281/zenodo.22085054). The version-controlled source code, tests, frozen protocol implementations, and figure-generation scripts are publicly available through the [tagged GitHub](#)

source [snapshot](#) linked from that Zenodo record. Zenodo is the citable authority for the saddle-node and marginal-tube result records; the immutable GitHub tag is the corresponding public code record.

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